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Recuperative potential of Indian medicinal plant compounds– a tool to encumber henipaviruses: an in –silico study

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Abstract

Henipaviruses, highly fatal zoonotic viruses with mortality rates up to 100%, pose a significant threat to humans. Despite sporadic cases, including infections from Cedar, Langya, and Nipah Viruses, there are no established drugs or vaccines for treatment. This lack of specific medication led us to explore 57 non-toxic compounds from Indian Medicinal Plants, selected from 232 compounds, aiming to combat these viruses. Through in silico ADMET analyses, Three compounds—andrographolide, pterygospermin and Salidroside—stood out for their exceptional non-toxic properties. These compounds underwent in silico target prediction, molecular docking and dynamics with Cedar, Langya, and Nipah Virus proteins from the Protein Data Bank. Among them, Andrographolide displayed the most

promising negative free energy scores and stability in Cedar Virus-Attachment G-Protein binding pockets. Pterygospermin and Salidroside showed efficacy against Langya and Nipah Virus target proteins throughout the simulation. These compounds not only exhibited antiviral properties but also demonstrated immunomodulatory, anti-inflammatory, and hepatoprotective effects by our in-silico studies. Their potential as treatments or preventive measures against henipaviral infections makes them promising candidates for further research and development.

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Data availability

No datasets were generated or analysed during the current study.

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